Business Statistics

Visualizing Categorical Variables

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

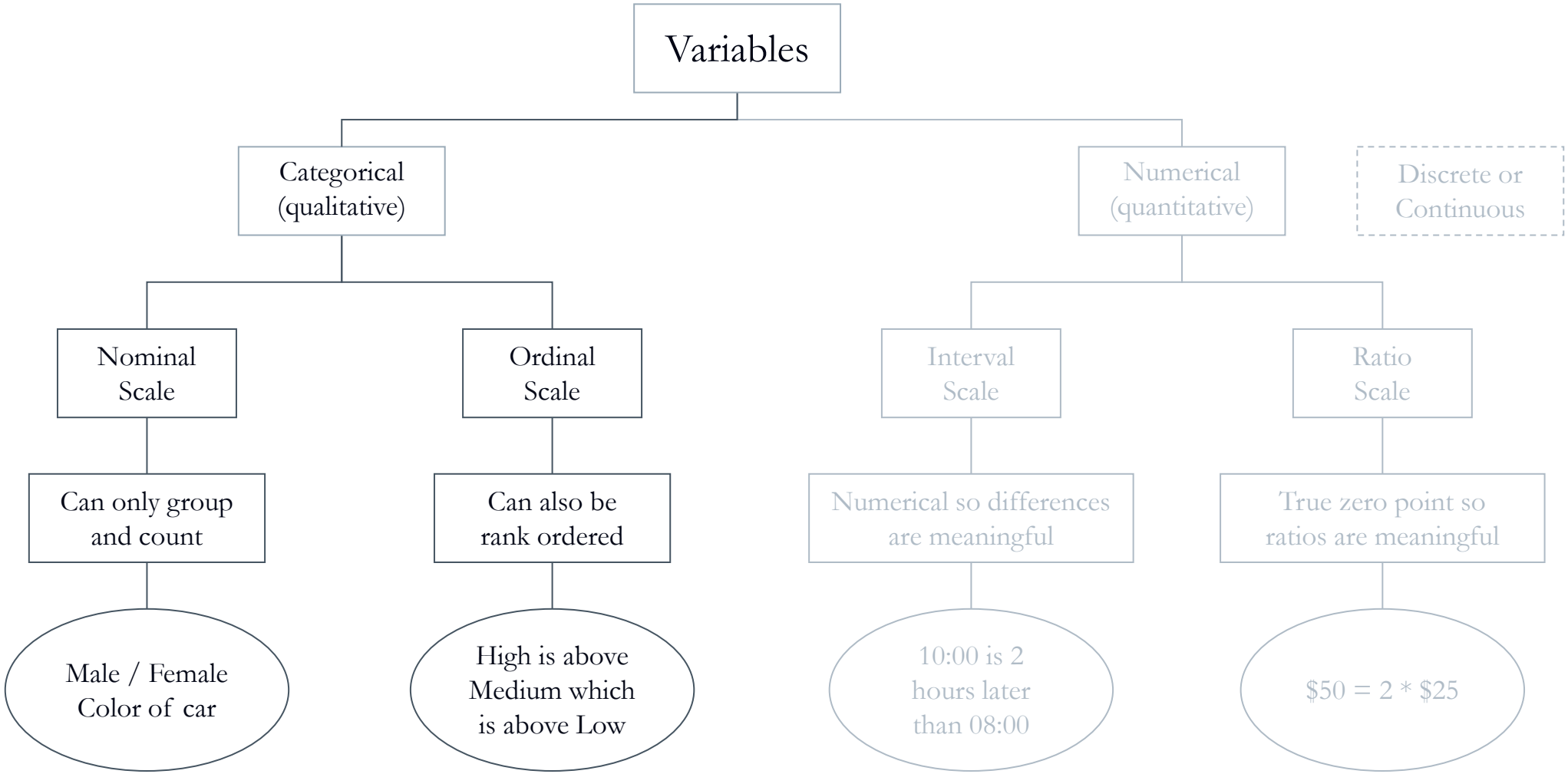
Visualizing Categorical Variables Overview

- Nominal and ordinal variable types

- Visualizing one categorical variable
  - Preparing the data: frequency distributions
  - Bar and Pareto Charts
  - Pie Charts

- Visualizing two categorical variables
  - Preparing the data: contingency tables
  - Clustered Column Charts
  - Stacked Column Charts

Categorical data
Nominal or ordinal?



Visualizing one categorical variable

Preparing the Data: Frequency Distribution



Visualizing Categorical Variables
Preparing data: frequency distribution

- Raw data:
  - List of 30 products purchased
  - Nominal data

Product bought
Electronics
Clothing
Groceries
Electronics
Home Goods
…
Sports
Groceries
Electronics

- Group and create frequencies
  1. Group by category
  2. Count frequency by category
  3. Get relative frequencies by category
  4. Get percent frequencies by category

Category	Frequency	Relative Frequency	Percent Frequency
Clothing	6	0.20	20%
Electronics	10	0.33	33%
Groceries	8	0.27	27%
Home Goods	3	0.10	10%
Sports	3	0.10	10%
5	30	1.00	100%

Visualizing Categorical Variables
Preparing data: cumulative frequency distribution

- Cumulative frequency is based on the ordering of the data

- Here the categories are sorted alphabetically
  - Cumulative frequencies here are meaningless and misleading

- Here the categories are sorted by frequency from high to low.
  - Cumulative frequencies here give useful information
  - It shows the percent of category purchases from high to low

Category	Frequency	Relative Frequency	Percent Frequency	Cumulative Frequency
Clothing	6	0.20	20%	20%
Electronics	10	0.33	33%	53%
Groceries	8	0.27	27%	80%
Home Goods	3	0.10	10%	90%
Sports	3	0.10	10%	100%
5	30	1.00	100%	100%

Category	Frequency	Relative Frequency	Percent Frequency	Cumulative Frequency
Electronics	10	0.33	33%	33%
Groceries	8	0.27	27%	60%
Clothing	6	0.20	20%	80%
Home Goods	3	0.10	10%	90%
Sports	3	0.10	10%	100%
5	30	1.00	100%	100%

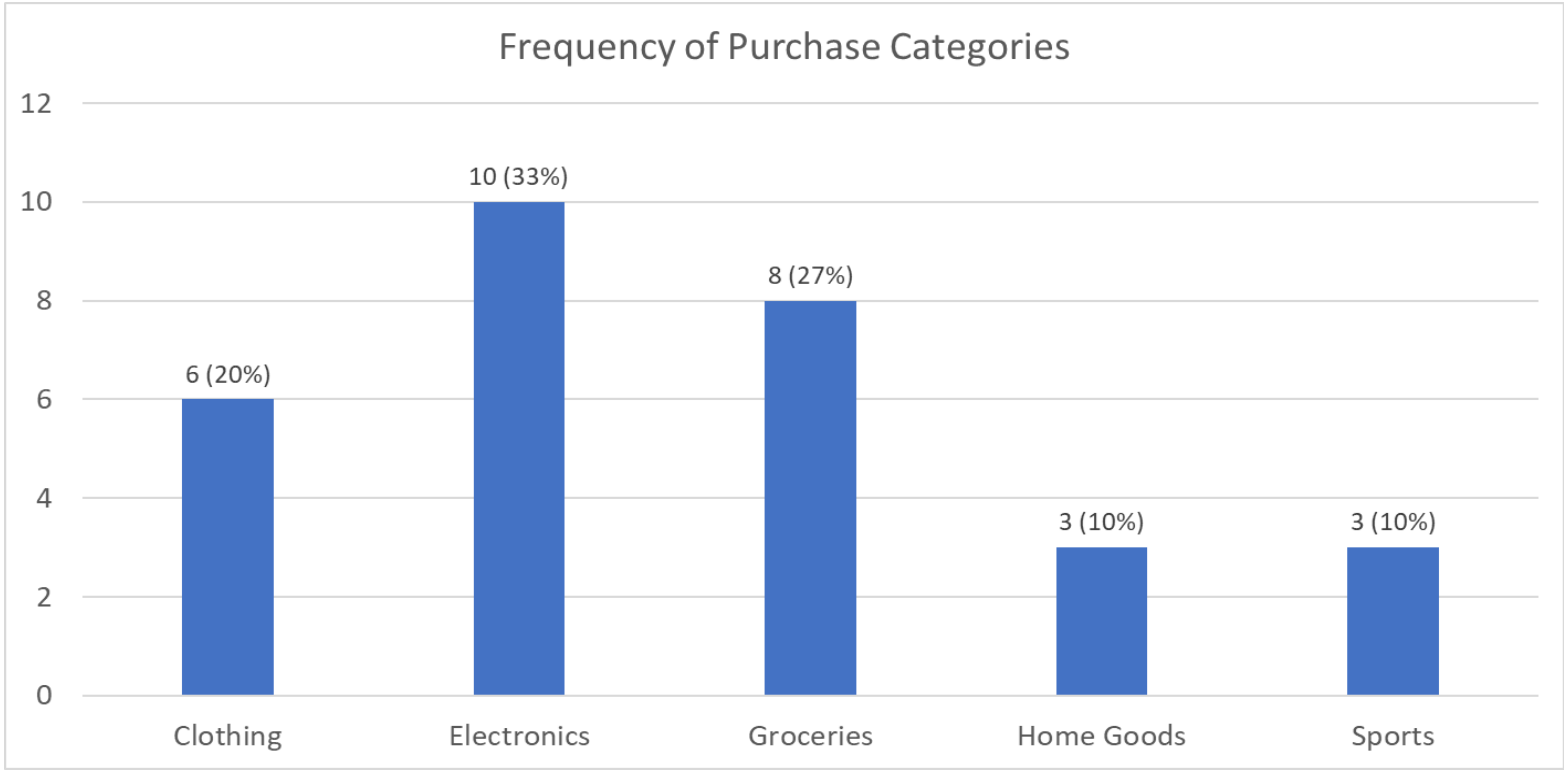
Visualizing one categorical variable

Bar and Pareto Charts



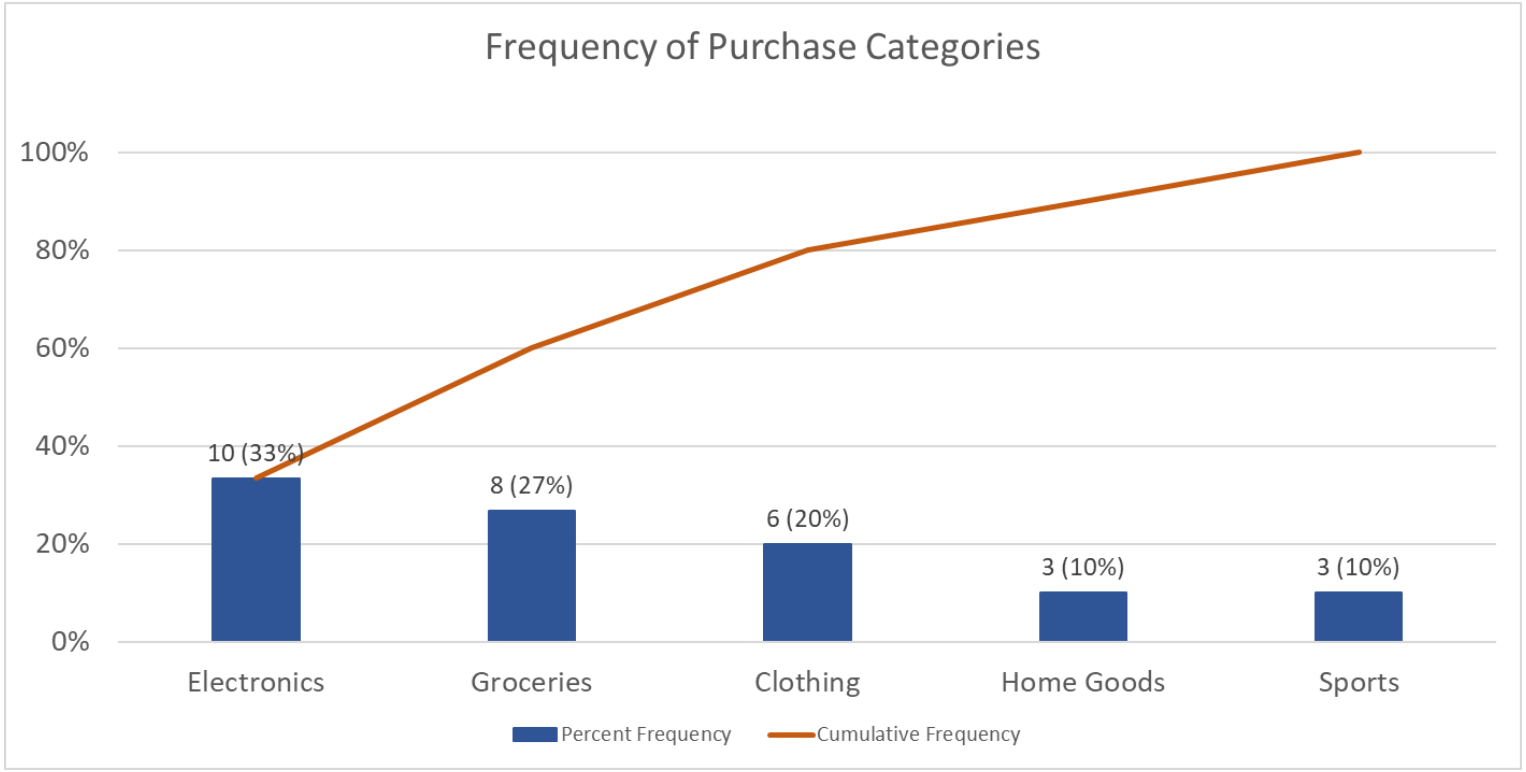
Visualizing Categorical Variables
Bar charts of the category frequencies

- Bar charts visually display the relative frequency of the categories
- The data here has been sorted alphabetically by category
- Custom data labels include absolute and percentage frequencies



Visualizing Categorical Variables
Pareto chart orders bars by frequency

- Pareto charts display the categories sorted by frequency, high to low
- Generally, you do **NOT** sort categories for ordinal data



Visualizing one categorical variable

Pie Chart

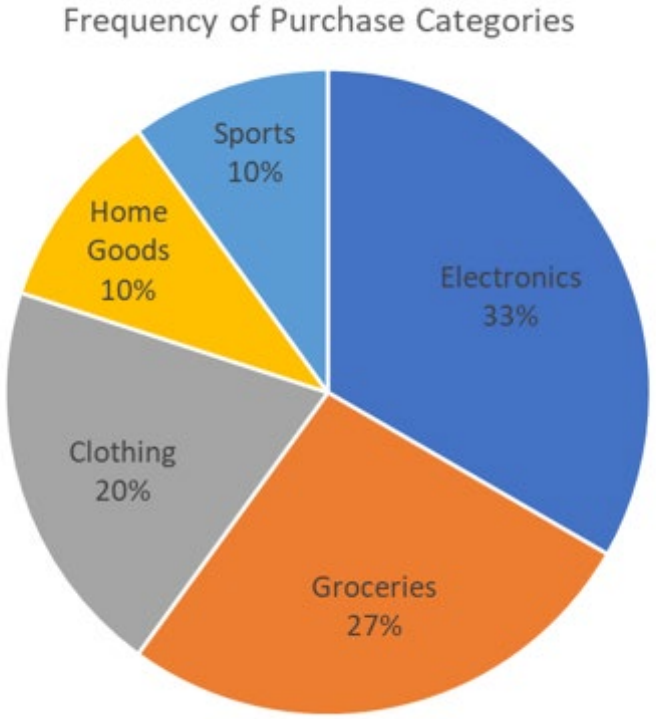


Visualizing Categorical Variables
Pie charts ordering of categories

- Nominal data:
  - If there is a natural order of categories, then use that natural order
  - E.g., always report stores in certain order, or regions, etc.
  - Otherwise, in general sort the categories by frequency

- Ordinal data:
  - Keep that order

- Comparing multiple charts:
  - When comparing multiple charts they should all be in the same order



Visualizing two categorical variables

Preparing the Data: Contingency Table



Visualizing Categorical Variables
Preparing data: contingency table

- Raw data:
  - 30 purchases by payment type
  - Nominal data
- Create cross-classifications by count and relative frequency

PurchaseID	PurchaseType	PaymentMethod
P001	Electronics	Credit Card
P002	Clothing	Debit Card
P003	Groceries	Cash
…	…	…
P028	Home Goods	Debit Card
P029	Groceries	Cash
P030	Clothing	Credit Card

	Credit Card	Debit Card	Cash	Marginal
Electronics	8	0	1	9
Clothing	1	5	3	9
Groceries	0	5	3	8
Home Goods	3	1	0	4
Marginal	12	11	7	30

	Credit Card	Debit Card	Cash	Marginal
Electronics	27%	-	3%	30%
Clothing	3%	17%	10%	30%
Groceries	-	17%	10%	27%
Home Goods	10%	3%	-	13%
Marginal	40%	37%	23%	100%

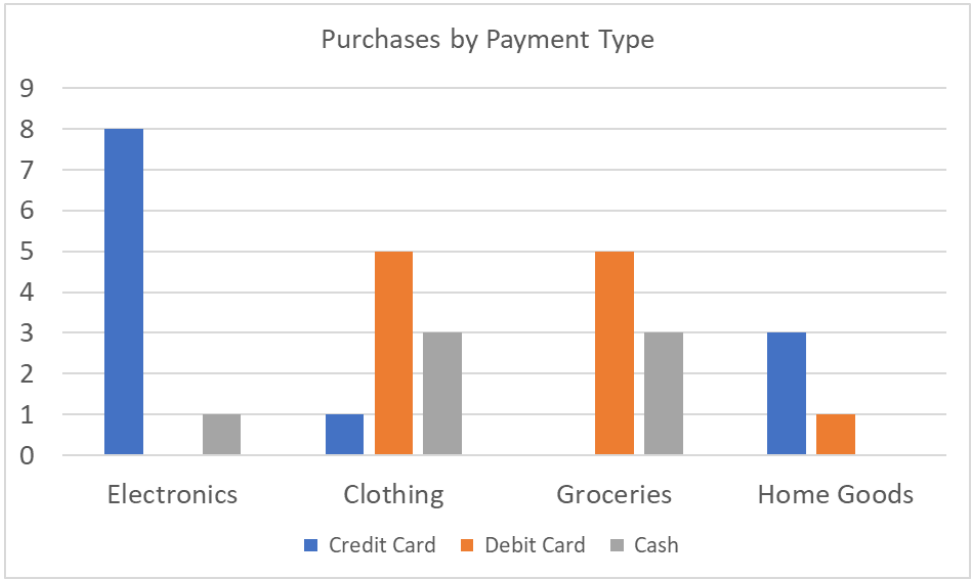
Visualizing two categorical variable

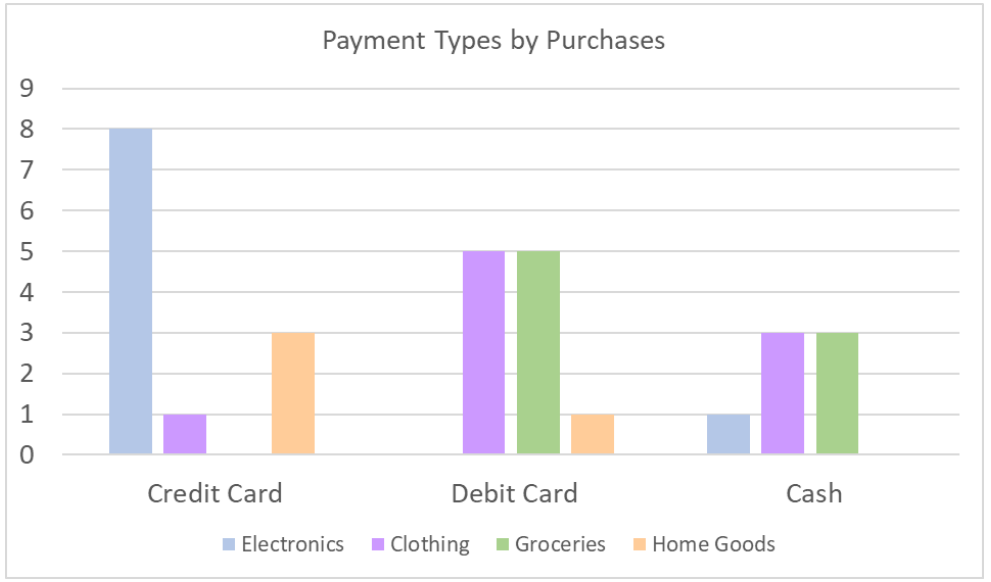
Clustered and Stacked Column Charts



Visualizing Categorical Variables
Clustered bar charts

- Clustered chart: **absolute** breakout of one variable's frequency by categories of the other variable

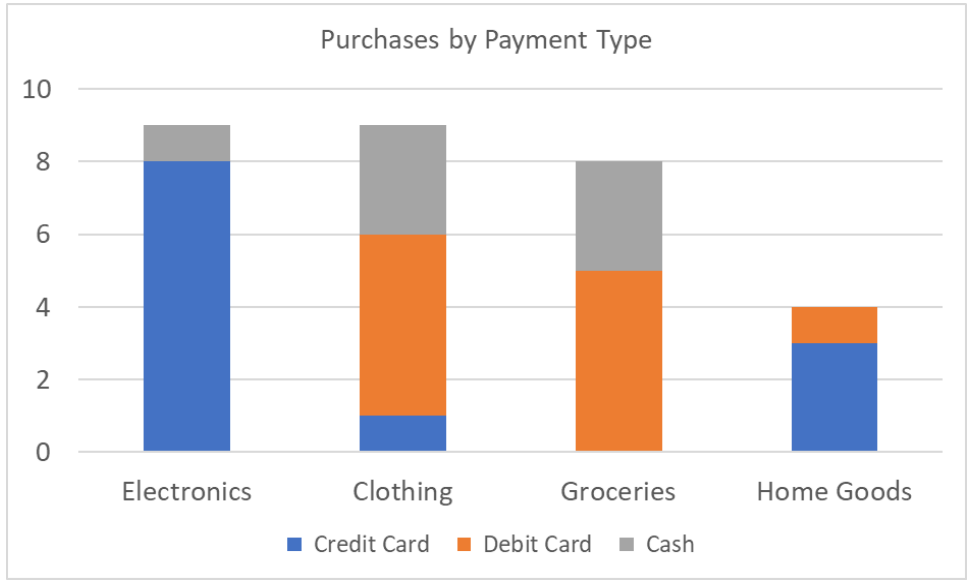


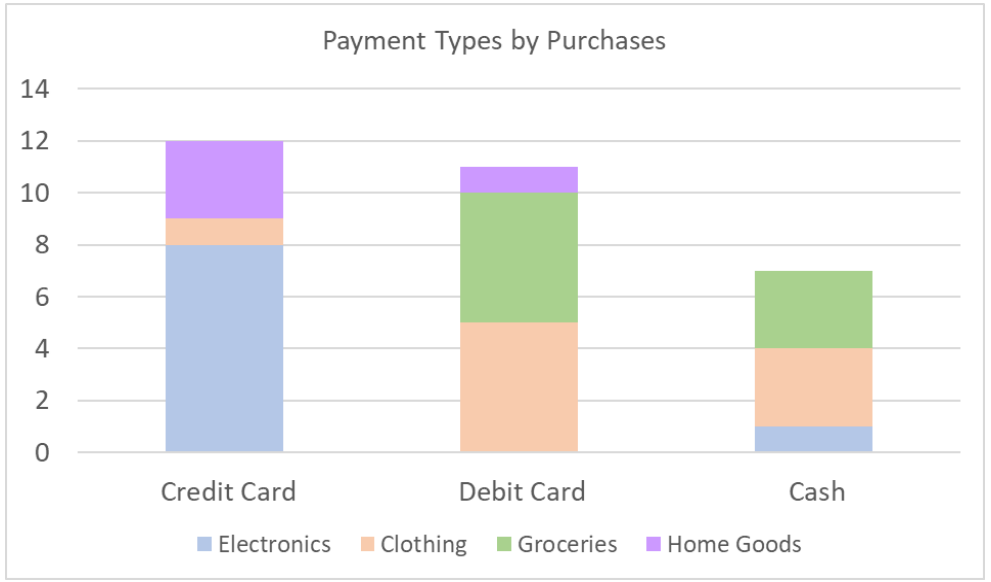


- To create one chart from the other, in Excel "Select Data" and then "Switch Row/Column", update title, and change series colors

Visualizing Categorical Variables
Stacked bar chart - absolute

- Stacked chart: **absolute** breakout of one variable's frequency by categories of the other variable, but stacked

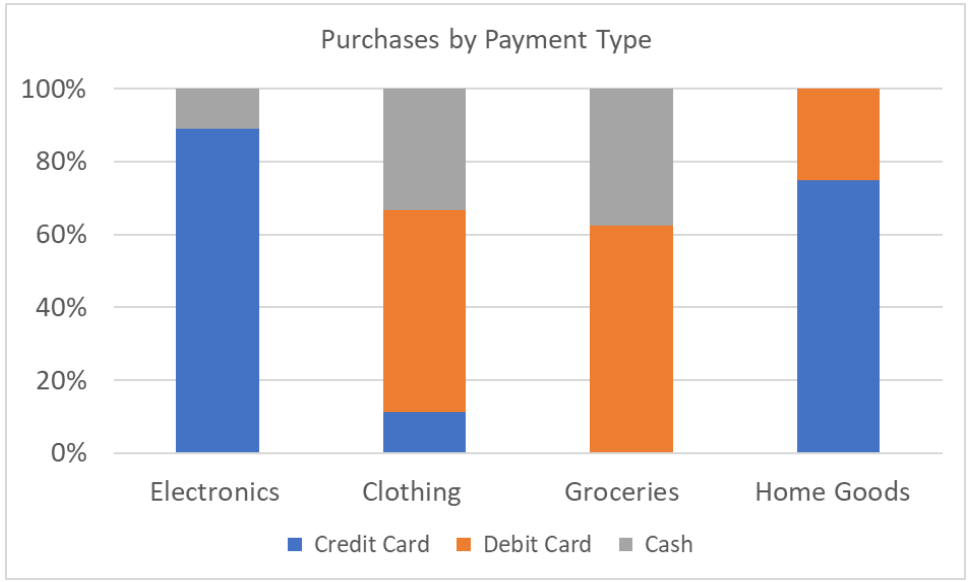


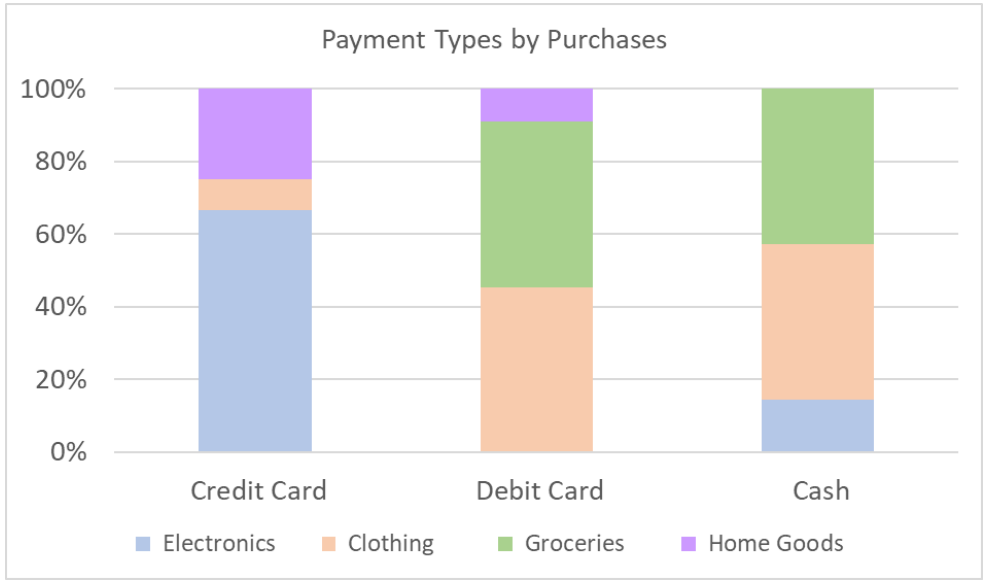


- To create one chart from the other, in Excel "Select Data" and then "Switch Row/Column", update title, and change series colors

Visualizing Categorical Variables
Stacked bar chart – 100%

- 100% Stacked chart: **proportional** breakout of one variable's frequency by categories of the other variable





- To create one chart from the other, in Excel "Select Data" and then "Switch Row/Column", update title, and change series colors